

Neeraj Jadhav

Education

- 2024-26 **Master of Science**, *Computer Science*, Specialization: Computing Systems
Master's Thesis: Can LLMs do program behavior analysis and optimization?
Prof. Santosh Pande, Georgia Institute of Technology, GPA: 4.0
- 2020-24 **Bachelor of Technology with Honors**, *Computer Science and Engineering*
Minor in Artificial Intelligence and Data Science
Indian Institute of Technology, Bombay, (Major) CGPA: 9.34, (Minor) CGPA: 9
Key courses: Algorithms, Operating Systems, Compilers

Publications

- Phaedrus: Exploring Dynamic Application Behavior with Lightweight Generative Models and Large-Language Models (Submitted to PLDI, 2025)
Bodhisatwa Chatterjee, Neeraj Jadhav, Sharjeel Khan, and Santosh Pande

Honors & Awards

- Recipient of **Masters Graduate Fellowship in CS** by *Georgia Institute of Technology* ['24]
- Secured **46th**/150k and **186th**/870k national ranks in **JEE Advanced** and **Main** exams resp. ['20]

Professional Experience

- Summer '23 **Quantitative Research Internship**, *Quadeye Securities LLP*, Gurugram, India
Engaged in **large-scale** global optimization of **trading strategies** using **Bayesian** Optimization, **Genetic & Whale** Algorithms achieving **80% reduction** in time spent and **2% increased PnL**
- Winter '22 **Software Engineering Internship**, *FinIQ Consulting*, Pune, India
Enhanced the proprietary financial coding language **FinSPL**, by adding **operational parallelism** and **efficient** data structures for information **storage** and **retrieval** of streaming **market indices**
- Summer '22 **Automated Orchestration using LLMs**, *IBM Research*, Gurgaon, India
Streamlined business task execution using **LLMs** and **process mining** to optimize **workflow efficiency**
Pipelined **data extraction** from **conversation logs** by fine-tuning **GPT2** for seamless **API calls**

Research Experience

- Fall '23 **Robust Celltree : A Paradigm for Distributed Data Repositories**, *Bachelor's Thesis*
Prof. Manoj Prabhakaran (IIT Bombay), Prof. Indranil Gupta (UIUC)
Designed a **new data structure** with **safety** guarantees to **minimize** computational **load and time**
Enhanced **blockchain** with **proof-of-stake** and modified **consensus**, achieving high **latency gains**
- Fall '23 **mpC: A Secure Multiparty Computation Language**
Prof. Manoj Prabhakaran (IIT Bombay)
Developed a **language**, with backend in **C++**, for **abstracting** out **cryptographic protocols**
Secured **communication with TLS & reduced** compilation **time** for **rapid testing** of new protocols

Key Projects

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|------------|--|---|
| Summer '24 | Trading Simulator and Backtester | Personal Project |
| Spring '23 | ipLC : Miniature GCC-like C Compiler | Guide: <i>Prof. Amitabha Sanyal (IITB)</i> |
| Spring '23 | Blockchain Simulator | Guide: <i>Prof. Vinay Ribeiro (IITB)</i> |
| Spring '22 | Peer-to-Peer Networked File Sharing Application | Guide: <i>Prof. Kameswari Chebrolu (IITB)</i> |
| Spring '22 | RISC 16 Bit Processor in VHDL | Guide: <i>Prof. Virendra Singh (IITB)</i> |

Leadership

- Mentorships **Student Mentor & Tech Mentor at XLR8 Robotics competition**, *IIT Bombay*
TAsHips **Calculus & Embedded Systems**, *IIT Bombay*, **Algorithms**, *GATech*

Technical Skills

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|----------------------------------|---|
| Languages & Libraries | C/C++, Python, R, JavaScript, SQL, Bash, Huggingface, PyTorch |
| Software Frameworks | Git, MATLAB, Wireshark, Django, React.js, Node.js |

Extracurricular Activities

- Member of the **Campus Cats Club** at GATech, supporting **initiatives to care for** campus cats
- Amongst **top 10** teams in **Algorithmic Trading** contest organized by Web and Coding Club, IITB
- Engage in playing the **violin** to enhance musical proficiency, contributing to theatrical **performances**